

EMK Constructive Cut Foundation

Typed records, native operators, iota, recognition geometry and completion

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Corrective mathematical edition

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Abstract

We construct an explicit foundation from finite typed records of cut and self-cut. Cut counts supply signed refinement scalars. The two recorded roles supply parity and exchange maps whose composition has a derived four-cycle and square minus identity; no complex field or cyclic phase alphabet is an input. The complex-like scalar sector is subsequently recovered inside the larger operator algebra. Finite recognition depth supplies idempotent apertures, a prefix topology, a native ultrametric and a complete history space. Full history remains distinct from operator or scalar shadow. Exact comparison-loop residues characterize path-independent increments.

This is a *corrected constructive theory*, not a proof of the original manuscript's unrestricted three-prose-axiom claim. The typed recursion and completion operations are stated explicitly. In particular, closure of the constructed role operators is not literal closure of arbitrary primitive acts. Every theorem in the new core has a written proof or an identified native analytic dependency. Exact finite certificates and source hashes accompany it; no proof-assistant verification or physical identification is claimed.

1 The source, and what has been corrected

The input manuscript `emk_topology.tex` has SHA-256

55f449e7eb3a2bde9e77193f595a6a5eeff3923da2bea076d00f0e9907b8cebd.

It introduces a ground, a cut from the ground, and a self-cut on the trace of a cut. Later it applies both acts recursively to all appearances, identifies their product with an involution, and derives further structures from that identification. Those are different mathematical assertions.

The present source is the *free typed realization of the stated recursion*. There are sorts Ground, History and Trace. A first-cut constructor gives a seed p_0 from the ground. There is a history-extension constructor k ; recording a history gives its trace; a self-cut applied to that trace gives the constructor c . Thus subsequent histories are finite words in the two event tags k, c , based at p_0 .

The history with no *additional* event is denoted ϵ . It is the post-cut seed p_0 , not the pre-cut ground and not absence of an object. Zero additional cut count therefore does not identify the ground with a scalar zero. The map from histories to their count is explicitly a summary, not history equality.

The extension and trace constructors make the manuscript's recursive domain use precise. They are a stated formalization, not a theorem that its original prose uniquely forces this formalization. The proof language uses logical equality, finite records, functions, subsets and induction. These are declared metamathematical infrastructure. No ordinary complex numbers, spacetime, metric, noise distribution or physical clock is an initial object.

Theorem 1.1 (C1: Free record and evaluation theorem). *Finite based records form a free monoid under concatenation. Every assignment of the two tags to composable endomaps has a unique evaluation preserving concatenation and the empty record. For general typed arrows, the analogous construction has separate local identity paths and composition only at matching endpoints.*

Proof. The empty record must evaluate to the appropriate identity. Removing the last tag of a nonempty record leaves a shorter record; its value and the value of the final tag determine the evaluation. Induction gives existence and uniqueness. Concatenation is associative because either parenthesization gives the same ordered list. Typed endpoint matching proves the same statement object by object. $\square$

Proposition 1.2 (C2: What primitive naming does not prove). *The free record $ckck$ is not the empty record. The equation $(\chi\kappa)^2 = I$ is not a consequence of specifying two composable acts.*

Proof. The two records have different lengths. Alternatively, take three appearances and maps $\kappa = (1, 2, 0)$, $\chi = (0, 1, 2)$ in table notation. The composite squared sends 0 to 2, not 0. A separate initial cut can still create the seed 0. Thus even the totalized algebraic reading permits failure of the claimed equation. $\square$

The result below derives closure on a *constructed role carrier*. It does not erase this counterexample or silently reinterpret a literal primitive identity. This distinction is part of the certificate.

2 Counts, signs and refinement before scalar coordinates

Let a cut count be the length of a finite string of identical counting marks. The empty counting string has count zero and appending a mark is successor. Disjoint concatenation defines addition; replacing each mark by a copy of a second counting string defines multiplication.

Theorem 2.1 (C3: Signed and refined cut counts). *Signed pairs (p, n) modulo*

$$(p, n) \sim (p', n') \iff p + n' = p' + n$$

form a commutative ordered ring of signed cut counts. Refinement packets $[a : b]_\Sigma$, with signed a and positive count b , modulo $ab' = a'b$, form an ordered field $\mathbb{Q}_\Sigma$ under

$$[a : b] + [u : v] = [av + ub : bv], \quad [a : b][u : v] = [au : bv].$$

Proof. Concatenation and rectangular replication give associative, commutative count addition and multiplication, with distributivity, by matching finite positions. Count cancellation follows by removing equal initial blocks, and induction compares any two counts. Adding signed pairs componentwise and setting $(p, n)(u, v) = (pu + nv, pv + nu)$ respects the displayed relation by expansion. The swapped pair is the additive inverse. The sign of the difference is well-defined by count comparison. A product of two positive nonzero differences is positive, so there are no zero divisors.

Cross-multiplication shows the refinement relation is reflexive and symmetric; transitivity uses cancellation of its positive common denominator. Expanding the two operation formulas shows independence of representatives. The field laws follow by putting finitely many operands over a common positive denominator and using the signed ring laws. For nonzero $[a : b]$, interchange numerator and denominator and move any denominator sign to the numerator; the resulting product is one. Cross-multiplied signs give the field order. $\square$

Opposite signs cancel in the *coefficient ledger*. They do not delete the underlying cut-event record or physically undo an irreversible cut. This is the native count/refinement content credited to MP NR-7 in the source ledger.

3 A four-cycle derived from the two cut roles

Form the free signed ledger on the two distinct recorded roles e_k, e_c . After refinement, denote it $V = \mathbb{Q}_\Sigma e_k \oplus \mathbb{Q}_\Sigma e_c$. This is an additive construction on tags; its elements are not asserted to be all UGD states.

Role exchange and role parity are the following explicitly constructed maps:

$$Ke_k = e_c, \quad Ke_c = e_k, \quad He_k = e_k, \quad He_c = -e_c.$$

Exchange is induced by the swap of the two tags. Parity takes the additive inverse of the self-cut-role count, while leaving the cut-role count fixed. Their extension is unique by additivity. The order of the named roles fixes the displayed orientation convention; reversing it reverses the quarter-turn.

Theorem 3.1 (C4: Native role-operator closure). *Define $R = KH$. Then, on every signed/refined role ledger,*

$$H^2 = K^2 = I, \quad R^2 = -I, \quad R^4 = I, \quad KR = -RK.$$

The orbit of e_k under R is precisely $e_k, e_c, -e_k, -e_c, e_k$. The group generated by H, K consists of the eight signed permutations $\pm I, \pm H, \pm K, \pm R$.

Proof. Directly on the two free generators, $Re_k = e_c$ and $Re_c = -e_k$. Applying twice negates each generator, and applying four times fixes each. Exchange and parity each square to identity. Also $KRe_k = e_k = -RKe_k$ and $KRe_c = -e_c = -RKe_c$, proving anticommutation. Additivity extends each identity to every ledger.

All products reduce to the eight listed signed permutations using these relations. Their images of the two generators are pairwise distinct; each listed permutation is generated by H, K . Thus the list is exact. $\square$

Here the number four is an *output*, not a supplied phase-alphabet size. The constructed maps are definitions on the free role completion; the closure equations are their proved consequences. No theorem has identified these maps with every possible physical realization of primitive acts.

The free evaluation of Theorem 1.1 permits the assignment $k \mapsto H, c \mapsto K$. Its image of the composite ck is R . Under this particular interpretation $(ck)^2$ acts as $-I$, and $(ck)^4$ acts as I . Four quarter-turns therefore mean eight alternating primitive-tag occurrences here. They are not the upload's asserted four primitive-tag identity. This correction prevents an off-by-two cycle claim.

4 The larger operator algebra and its scalar sector

Theorem 4.1 (C5: Algebra and recovered scalar sector). *The algebra generated by the role operators is all $\text{End}_{\mathbb{Q}_\Sigma}(V)$, with basis I, K, R, RK . The operators commuting with R are precisely*

$$\mathbb{K}_\Sigma^0 = \{aI + bR : a, b \in \mathbb{Q}_\Sigma\}.$$

This commutative sector is a field, with derived element $\iota = R$,

$$\iota^2 = -1, \quad (a + b\iota)(c + d\iota) = (ac - bd) + (ad + bc)\iota, \quad (a + b\iota)^{-1} = \frac{a - b\iota}{a^2 + b^2}$$

for nonzero $a + b\iota$.

Proof. Read the two generator images after constructing the operators. Their output coordinate arrays are

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad RK = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

They span all four independent array positions because division by two exists in $\mathbb{Q}_\Sigma$. Comparing the two images in $AR = RA$ forces an arbitrary array A to have equal diagonal entries and opposite off-diagonal entries, hence $A = aI + bR$. Its product law follows from $R^2 = -I$. It is closed and commutative. The positive ordered-cut expression $a^2 + b^2$ vanishes only when both coefficients vanish. Multiplying by the displayed inverse gives identity. Associativity and distributivity come from composition and addition of the constructed maps. $\square$

These arrays are outputs of the free-generator calculation. They were not assumed as primitive geometry. Further, R is central in $\mathbb{K}_\Sigma^0$ but not in the full algebra: it anticommutes with K . The scalar sector therefore cannot replace the noncommuting algebra or the full UGD state.

5 Pairing, dagger, seam and operator curvature

Coefficient matching of the two recorded roles defines

$$B(ae_k + be_c, ue_k + ve_c) = au + bv, \quad \omega(ae_k + be_c, ue_k + ve_c) = av - bu.$$

Theorem 5.1 (C6: Native pairing and compatible orientation). *B is bilinear, symmetric and positive definite. It is preserved by H, K, R . The oriented form satisfies $\omega(x, y) = B(Rx, y)$. Within the scalar sector, orientation reversal gives*

$$(a + bu)^\dagger = a - bu, \quad z^\dagger z = a^2 + b^2, \quad N(zw) = N(z)N(w).$$

Proof. Bilinearity and symmetry follow by distributivity. Its diagonal is the sum of the two signed-cut squares, positive for a nonzero ledger. Sign changes and tag exchange preserve matched products, hence so does their composite. Substitution of $Rx = -be_k + ae_c$ gives the oriented-form identity. Reversing the quarter-turn changes R to $-R$; expansion proves the dagger laws and the norm-square identity. Expanding the product norm gives $(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2$. $\square$

The pairing is derived from this marked-role counting construction. It is not declared to be a universal physical spacetime metric, nor automatically the pairing on every future UGD or morphic module.

Theorem 5.2 (C7: Seam aperture and genuine mixed curvature). *The fixed seam of K is $\mathbb{Q}_\Sigma(e_k + e_c)$, including both signs. Put $P = (I + K)/2$ and $Q = (I - K)/2$. Then*

$$P^2 = P, \quad Q^2 = Q, \quad PQ = QP = 0, \quad P + Q = I.$$

The native mixed order defect is $[R, K] = 2RK \neq 0$, whereas $[K, K] = 0$. For any two endomaps A, B of V ,

$$PABP - (PAP)(PBP) = PAQBP.$$

Proof. Exchange fixes exactly equal role coefficients. Expand the projector expressions using $K^2 = I$. Anticommutation gives the mixed defect; its value on e_k is $-2e_k$, which is nonzero. The final identity follows by inserting $I = P + Q$ between A and B . $\square$

The last formula credits the existing native cut-corner theorem. It identifies the returning omitted component algebraically, without assigning a probability law or physical noise mechanism. That later work remains separate.

6 Cut-Cauchy scalar completion and native analytic dependencies

Define $D(a + bi) = |a| + |b|$ in the ordered cut field. Direct expansion proves the triangle inequality and $D(zw) \leq D(z)D(w)$.

Theorem 6.1 (C8: Complete native coefficient field). *Cut-Cauchy sequences in $\mathbb{K}_\Sigma^0$, modulo differences tending to zero in every positive rational-cut radius for D , form a complete field $\mathbb{K}_\Sigma$. Termwise addition, multiplication and dagger are well-defined, and the constant-sequence map embeds $\mathbb{K}_\Sigma^0$ densely. The radial component is the ordered Archimedean completion $\mathbb{R}_\Sigma$ of $\mathbb{Q}_\Sigma$. Each nonnegative radial scalar has a unique nonnegative square root, and $\sqrt{N(z)}$ is a multiplicative norm defining the same completion.*

Proof. A Cauchy sequence is bounded: after its Cauchy threshold it stays within one unit of a fixed term, and finitely many earlier terms have a common count bound. The two inequalities above show that sums and products of Cauchy sequences are Cauchy and that changing either by a null sequence changes the result by a null sequence. Dagger preserves D .

A nonzero Cauchy class has some positive rational ϵ for which arbitrarily late terms have $D \geq \epsilon$. Choose a Cauchy tail whose pairwise differences have $D < \epsilon/2$. Comparison with a later large term shows every term of that tail has $D \geq \epsilon/2$. Write this lower bound as δ . Since $a^2 + b^2 \geq (|a| + |b|)^2/2$, the inverse formula gives $D(z^{-1}) \leq 2/\delta$. Consequently

$$D(z_n^{-1} - z_m^{-1}) \leq (4/\delta^2)D(z_n - z_m).$$

Tail inverses are Cauchy and give the inverse class; finitely many initial choices do not affect it.

Every class is approximated by sufficiently late individual terms of a representative, so constant cut scalars are dense. For completeness, approximate the j th member of a Cauchy sequence of classes by a rational-cut element within 2^{-j} . Enumerate rational-cut pairs by increasing total numerator/denominator count, breaking ties by the ordered role tags. Density supplies an eligible element; select the first eligible element of this fixed enumeration. Here radius comparison between classes means eventual comparison of representative differences, with strict rational slack, so it is independent of representatives. No simultaneous choice of representatives is required. The approximating elements are Cauchy by the triangle inequality and their class is the limit. Constant sequences are injective because a nonzero exact cut scalar has a positive rational separation from zero.

Radial sequences remain radial under these operations. Their nonnegative cone is the closure of the nonnegative rational cuts; comparison by rational separation gives a proper total order. A Cauchy representative is bounded by a finite cut count, proving the Archimedean property. Componentwise construction identifies the full completed field with two radial components and the already derived multiplication law, without taking ordinary $\mathbb{C}$ as input.

For a nonnegative radial s , choose a count $M > s + 1$. Bisect $[0, M]$, keeping the half whose squared endpoints bracket s . The nested endpoints are Cauchy because their separation is $M2^{-n}$. Their common limit u satisfies $u^2 = s$, since products respect limits. If $0 \leq u < v$, then $v^2 - u^2 = (v - u)(v + u) > 0$, proving uniqueness. Thus square root is constructed, not imported. Finally $(ac + bd)^2 \leq (a^2 + b^2)(c^2 + d^2)$ follows by subtracting to obtain $(ad - bc)^2 \geq 0$. Expanding $N(z + w)$ and taking the nonnegative roots gives the triangle inequality for $\sqrt{N}$. Positivity and multiplicativity follow from Theorem 5.1. The inequalities $N \leq D^2 \leq 2N$ prove agreement of their Cauchy notions. $\square$

This is the constructive content of MP NR-7/NR-8; the original oriented-plane input has here been supplied by Theorem 3.1. Norm-square and D neighborhoods agree because $N \leq D^2 \leq 2N$, so their completions coincide.

The following *native analytic results are cited, not re-proved or presented as a new formal certification here*: RKF F00/E (N1) constructs the factorial-series exponential and Euler identities on this completed field; F00G (N2) constructs the positive logarithm and proves its product law; RH F00J (N3) constructs the fundamental circular period, and F00K (N4) proves scalar polar decomposition and the branch logarithm. Their exact files, commits and premises are in the source ledger. The coefficient and completion inputs are supplied above. They do not select a physical angle trajectory or make a scalar logarithm single-valued on full loop histories.

For clarity, the radial monotonicity input of F00G needs no additional classical calculus result: for $t > 0$ the native exponential series gives $\text{Exp}_\Sigma(t) > 1$. The exponential group law gives positivity also for negative t by reciprocation. For $y > x$, it then gives $\text{Exp}_\Sigma(y) = \text{Exp}_\Sigma(x) \text{Exp}_\Sigma(y - x) > \text{Exp}_\Sigma(x)$.

7 Recognition depth, topology and a native history metric

For a finite history h , read its two-tag word followed by an end marker $\perp$, repeated after termination. The marker is a record of termination, not the pre-distinction ground. Recognition at depth n reads the first n symbols of this padded record. Two histories are fully recognized as equal exactly when every finite readout agrees.

Theorem 7.1 (C9: Recognition equality and prefix topology). *Full finite-depth recognition is equality of histories. Sets fixing a finite readout form a clopen basis for a topology τ_{rec} . This is the coarsest topology in which every finite recognition map into its finite discrete record set is continuous.*

Proof. Unequal words differ at a symbol or at the first termination of the shorter word, so some finite readout separates them. A finite readout cylinder is partitioned by the next-symbol cylinders. Intersections of compatible cylinders are the longer cylinder; incompatible intersections are empty. This proves the basis laws. The complement of a cylinder is a union of the other cylinders at that depth, so it is clopen. Any topology making those finite maps continuous contains every cylinder; their union/intersection closure proves the claimed minimality. $\square$

On finite histories this topology is discrete: reading through a finite history's end marker isolates it. The same cylinders have a nontrivial boundary topology on completed infinite histories below. The forward-invariant topology of cut reachability is another construction and must not be identified with this recognition topology.

Theorem 7.2 (C10: Derived prefix ultrametric). *Let $\ell(h, g)$ be the length of the common prefix before the first differing tag or end marker. Set*

$$d(h, g) = \begin{cases} 0 & h = g, \\ 2^{-\ell(h, g)} & h \neq g. \end{cases}$$

Then d is a metric with the stronger triangle inequality

$$d(h, u) \leq \max\{d(h, g), d(g, u)\}.$$

It generates τ_{rec} . Prepending the same cut halves nonzero distance; appending the same cut is non-expansive.

Proof. Symmetry and separation follow from first difference. If each of two pairs agrees on its first n symbols, so does the third pair. Hence $\ell(h, u) \geq \min(\ell(h, g), \ell(g, u))$, which is the displayed inequality. A ball $d(h, g) < 2^{-n}$ fixes symbols through index n . Conversely every finite cylinder

contains a ball at each of its members. Thus the topologies agree. Prepending adds one to the common-prefix length. Appending cannot destroy a previously matching prefix, including when one original word ends first; hence it cannot increase distance. $\square$

The factor $1/2$ uses the native halving refinement of cut counts; it is an explicit normalization, not a number forced uniquely by the tag alphabet. More generally any rational $0 < r < 1$ in place of $1/2$ yields the same prefix topology. The numerical calibration is therefore a construction convention, not a selected physical length scale. This metric measures recorded distinguishability; it is not claimed to be Riemannian curvature geometry.

8 History completion, Eye apertures and the tower

Adjoin infinite two-tag histories to the finite histories. A completed record either has no end marker, or from its first end marker onward has only end markers. Extend the prefix metric by the same first-difference rule.

Theorem 8.1 (C11: Exact history completion). *The resulting space $\overline{\mathcal{H}}$ is complete, and finite histories are dense. It is the metric completion of the finite source records.*

Proof. In a Cauchy sequence, each finite prefix eventually stabilizes: take a radius smaller than the corresponding dyadic separation. The stabilized prefixes are compatible. If an end marker occurs, no later symbol can be a cut tag, because every sufficiently late original record obeys this finite condition. They therefore specify exactly one permitted finite or infinite history. Prefix stabilization is precisely convergence to it. Every infinite history is approximated within 2^{-n} by its finite length- n truncation; finite ones already belong to the dense source. The same description shows that equivalent Cauchy sequences have the same limit profile, establishing the completion identification directly. $\square$

Theorem 8.2 (C12: A correctly typed Eye hierarchy). *Let $E_n : \overline{\mathcal{H}} \rightarrow \overline{\mathcal{H}}$ truncate a history after at most n tags, retaining the result as an actual finite record. Then*

$$E_n^2 = E_n, \quad E_m E_n = E_{\min(m,n)}, \quad d(E_n h, h) \leq 2^{-n}.$$

Every history is recovered uniquely from its coherent truncations. Each E_n is non-expansive. There is no finite n that recovers every history.

Proof. Truncating twice keeps the shorter limit, proving the two algebraic identities. The truncated and original records agree in the first n positions unless already equal, proving the error bound. Identical coherent truncations fix every symbol, hence the whole history; the bound proves convergence back to that history. Truncation preserves any common prefix up to its cutoff and otherwise makes the results equal, proving non-expansiveness. Histories k^n and $k^n c$ have equal E_n and are distinct. $\square$

These are endomaps with a common domain, so idempotence is genuinely typed. For $m > n$, $E_m \neq E_n$ globally, as witnessed by a sufficiently long history. A particular short history may stabilize. Thus the global hierarchy can be strict without falsely requiring every individual state to change at every level. It is a refinement tower, not powers of one involution.

9 State return, history return and lifted phase

Theorem 9.1 (C13: No loss of cut history at operator return). *An evaluated operator word can be identity while its based event record is nonempty. For the derived quarter-turn, four R occurrences return every role ledger but leave four operation records. For n positive quarter-turn occurrences there is a unique decomposition $n = 4w + r$, $0 \leq r < 4$, with composition law*

$$(r, w)(s, v) = ((r + s) \bmod 4, w + v + \lfloor (r + s)/4 \rfloor).$$

Proof. The role return follows from $R^4 = I$; a four-letter word is not the empty word by Theorem 1.1. Repeatedly remove complete four-mark blocks to obtain the unique remainder and count. Adding two counts gives the formula, whose associativity is inherited from count addition. Forgetting w sends the counts zero and four to the same phase but erases their difference. $\square$

The full word contains more than its loop count. UGD phase/scale/seam objects must retain their active memory rather than be replaced by the scalar sector of Theorem 4.1. MP NR-9 proves the appropriate scalar-shadow quotient for its declared UGD law; it does not equate that quotient with the full UGD state. No identification of all history words with a single seam integer is made here.

Cut count gives an internal event index. Histories form a prefix order in which extension increases that count. Different branches need not be ordered as one physical time line. Neither event count nor a finite endpoint orbit establishes a continuous physical clock or a helical spacetime manifold.

10 Complete path-equality criterion from comparison residues

Let a finite cut graph have oriented edges and a signed/refined increment a_e on each edge. Formal reversed traversal carries $-a_e$. Reversal is a *comparison of records*, not a physical reverse operation that erases a cut. Choose a spanning forest by exploring finite paths. Put a potential F at zero on each root and integrate increments along the unique tree paths.

Theorem 10.1 (C14: Exactness and the full loop ledger). *For every edge $e : x \rightarrow y$ define $r_e = a_e - (F(y) - F(x))$. Tree-edge residues vanish. The following are equivalent:*

- (i) *every edge increment is a potential difference;*
- (ii) *the increment sum of every closed comparison walk is zero;*
- (iii) *every non-tree residue r_e is zero.*

There are $|E| - |V| + b$ independent fundamental comparison cycles, where b is the number of connected components. Same endpoints alone do not force equal increments.

Proof. Potential differences telescope on every walk, proving (i) implies (ii). The tree path from a root to x , followed by e , then the reversed tree path from y , has sum r_e , proving (ii) implies (iii). By the definition of F , tree increments are already its differences, so (iii) gives (i).

Each non-tree edge yields one such cycle with coefficient one on that edge and zero on other non-tree edges; hence these cycles are independent. Subtract these cycles from any signed cycle to leave a cycle supported on a forest. At a leaf its incident coefficient must be zero. Repeated leaf removal proves the remainder zero. A forest has $|V| - b$ edges, giving the stated count. $\square$

This supplies a complete finite criterion, including global periods; checking only one local commutator is insufficient. A directed diamond has no directed cycle but can have unequal sums on its two routes. A four-edge phase loop with unit increments has lifted sum four despite endpoint return. In a later native logarithm chart the analogous loop period must likewise be retained. For

the graph actually generated by the eight signed role operators and left multiplication by H, K , there are eight vertices and sixteen directed edges, hence nine independent comparison cycles. Its potential basis and every chord defect are included in the certificate.

11 What the completed core does and does not certify

The new core consists of Theorems/Propositions C1–C14 above. Each has a stated construction and proof. Its mathematical inputs are the typed free cut-record realization, the declared proof language and the explicit completion operations. The analytic extensions in Section 6 are named existing native results with their own evidence levels.

The original claim that *arbitrary* acts named first-cut and self-cut force all later structures remains false or unproved as recorded in the claim ledger. The new construction is not logically interchangeable with that universal statement. In particular:

- the four-cycle is proved on the constructed signed role carrier;
- the source’s asserted raw $(\chi\kappa)^2 = I$ is not restored;
- topology and metric are the constructed recognition geometry, not an automatic thermodynamic manifold;
- Maxwell relations, Legendre potentials, Onsager response, physical event probabilities and an infinite helix are not certified from a graph cycle or operator notation;
- the full UGD seam state is not equated with a complex scalar;
- no RH, Yang–Mills continuum or physical-constant claim is added.

Removing an incorrect theorem from the active core is a documented correction, not proof of the original theorem. The original file and all 48 dispositions remain available alongside this edition. Thus *the corrected constructive core has a closed proof ledger; the entire original manuscript is not certified as written*. Physical and later extra-work connections are deferred.

12 Certificate and source ledger

Run `certificate/verify.py` with the pinned RKF checkout. The local checker exhausts the eight derived signed permutations, all 512 group triples and all 32 pairing basis cases. These basis checks cover the full linear identities, not only sampled coefficients. It also tests 63 finite histories, all 250047 ultrametric triples, truncation bounds, 270 exact potential assignments on four graph types, and negative controls. General infinite statements have the written proofs above; the checker does not pretend to enumerate all infinite histories or analytic points.

The unchanged canonical RKF engine supplies algebra normalization and replay. The F00/E and F00GHI certificates are rerun against original source pins; their analytical theorems are not thereby converted into proof-assistant certificates. No Lean/Coq runtime is available in this environment, and no formal-proof-assistant verification is claimed. Source hash, written proof, finite algorithmic evidence and external review remain distinct.

Primary sources, with premises attached:

1. MP, commit `feedfca094aa557086a612a0736aaffc07ab1ae4`.
`proof_lab/native_operator_algebra/`: NR-7 signed cut/refinement field; NR-8 cut-Cauchy completion; NR-9 UGD scalar-shadow quotient. Native algebra/completion results. NR-7 originally *defines* the oriented action before proving its square; this edition constructs it from role exchange and parity. Private source bodies are cited, not republished.
2. RKF, commit `3cc5a33b05c16d59c90994ddda69dedc0d392424`, `operator_foundation/`: native typed paths, exact cut arithmetic, normal-form compiler, faithful regular action and

cut-corner residue. Native presentation inputs; no primitive ordinary Hilbert space. Canonical engine source is reused unchanged, not copied into this packet.

3. Same RKF commit, F00/E and F00G: native factorial-series Euler construction and positive logarithm. Inputs: the ordered cut completion and derived iota. Their finite source certificates are replayed; continuous results remain supported by the source's written proofs.
4. RH, commit c588ded973a395b5fce37c82f670616160979a2e, F00J and F00K: native period and scalar polar/logarithm results. Inputs: native Euler functions and ordered completion. Written-source citation, no new standalone formal or numerical certificate asserted here.
5. MP NR-13 normalized UGD group: algebraic gauge normalization is exact for its declared potential cocycle. Its later ordinary Haar and L^1/L^2 extension invokes analytic group/measure machinery and Young-type bounds. Those *analytic imports are not dependencies of this core*.

Exact URLs, Git blobs, SHA-256 hashes and evidence roles are recorded in `certificate/SOURCE_PINS.json`. Hashing establishes source identity, not theorem truth.

A Disposition of every original claim

The machine-readable ledger retains original environment titles and line ranges. A replacement is a corrected statement, not retroactive approval of the original wording. The following compact table covers all 48 original theorem, lemma, proposition and corollary environments, including duplicates.

Claim	Disposition	Corrected scope
ET01	Typed replacement	Free typed constructors make recursion lawful; the upload did not supply their recursive domains. <i>C1</i> .
ET02	Recognition replacement	Equality of all finite cut-record observations is an explicit equivalence. Observer wording alone does not define it. <i>C9</i> .
ET03	Recognition replacement	Equality of all finite cut-record observations is an explicit equivalence. Observer wording alone does not define it. <i>C9</i> .
ET04	Recognition replacement	Equality of all finite cut-record observations is an explicit equivalence. Observer wording alone does not define it. <i>C9</i> .
ET05	Recognition replacement	Equality of all finite cut-record observations is an explicit equivalence. Observer wording alone does not define it. <i>C9</i> .
ET06	Typed Eye replacement	Truncations are endomaps of one completed history space and satisfy the proved aperture meet law. <i>C12</i> .
ET07	Seam corrected	The actual exchange fixed set is the whole signed diagonal; its projector is derived. The original zero-cost and positive-ray statements are not certified. <i>C7</i> .
ET08	Topology replaced	Recognition cylinders, their prefix metric and completion are constructed. They are not the upload's unsaturated image-stability topology. <i>C9, C10, C11</i> .
ET09	Event index only	Cut count is an internal record index; a physical clock is not derived. <i>C13</i> .
ET10	Role closure corrected	Exchange/parity produce a quarter-turn on the signed role carrier. Arbitrary raw chi/kappa closure and a four-point orbit of one involution do not follow. <i>C2, C4</i> .
ET11	Tower corrected	Distinct truncation maps give a strict global hierarchy; powers of an involution do not. Individual finite histories can stabilize. <i>C12</i> .
ET12	Completion constructed	Signed cut refinements and Cauchy completion construct scalars; arbitrary finite/infinite histories form a separate completed record space. Alternation alone is insufficient. <i>C3, C8, C11</i> .
ET13	Typed Eye replacement	Truncations are endomaps of one completed history space and satisfy the proved aperture meet law. <i>C12</i> .

Claim	Disposition	Corrected scope (continued)
ET14	Seam corrected	The actual exchange fixed set is the whole signed diagonal; its projector is derived. The original zero-cost and positive-ray statements are not certified. <i>C7</i> .
ET15	Topology replaced	Recognition cylinders, their prefix metric and completion are constructed. They are not the upload's unsaturated image-stability topology. <i>C9, C10, C11</i> .
ET16	Pairing and order split	A native role pairing is derived, but distinct role operators need not commute; the mixed defect is nonzero. <i>C4, C6, C7</i> .
ET17	Full residue criterion	Endpoint equality does not erase periods. Potential exactness is equivalent to vanishing on every fundamental comparison cycle. <i>C13, C14</i> .
ET18	Infinite sum withdrawn	The scalar/history completions and finite loop criterion are proved. They do not validate the unspecified infinite master-seam series. <i>C8, C11, C14</i> .
ET19	Native analytic citation	Euler, radial logarithm, period and scalar polar/logarithm use pinned native theorems. The upload's arbitrary analytic-class equivalence is withdrawn. <i>C4, C5, C8, N1, N2, N3, N4</i> .
ET20	Mixed defect corrected	The mixed operator commutator is nonzero; every self-commutator is zero. This does not identify it with Riemannian or connection curvature. <i>C7, C14</i> .
ET21	Full residue criterion	Endpoint equality does not erase periods. Potential exactness is equivalent to vanishing on every fundamental comparison cycle. <i>C13, C14</i> .
ET22	Native analytic citation	Euler, radial logarithm, period and scalar polar/logarithm use pinned native theorems. The upload's arbitrary analytic-class equivalence is withdrawn. <i>C4, C5, C8, N1, N2, N3, N4</i> .
ET23	Role closure corrected	Exchange/parity produce a quarter-turn on the signed role carrier. Arbitrary raw chi/kappa closure and a four-point orbit of one involution do not follow. <i>C2, C4</i> .
ET24	Full residue criterion	Endpoint equality does not erase periods. Potential exactness is equivalent to vanishing on every fundamental comparison cycle. <i>C13, C14</i> .
ET25	Infinite sum withdrawn	The scalar/history completions and finite loop criterion are proved. They do not validate the unspecified infinite master-seam series. <i>C8, C11, C14</i> .
ET26	Native analytic citation	Euler, radial logarithm, period and scalar polar/logarithm use pinned native theorems. The upload's arbitrary analytic-class equivalence is withdrawn. <i>C4, C5, C8, N1, N2, N3, N4</i> .
ET27	Topology replaced	Recognition cylinders, their prefix metric and completion are constructed. They are not the upload's unsaturated image-stability topology. <i>C9, C10, C11</i> .
ET28	Topology replaced	Recognition cylinders, their prefix metric and completion are constructed. They are not the upload's unsaturated image-stability topology. <i>C9, C10, C11</i> .
ET29	Typed maps retained	Ordinary typed composition is valid. New recognition is full record equality, and truncations retain their common codomain; no unspecified quotient descent is assumed. <i>C1, C9, C12</i> .
ET30	Typed maps retained	Ordinary typed composition is valid. New recognition is full record equality, and truncations retain their common codomain; no unspecified quotient descent is assumed. <i>C1, C9, C12</i> .
ET31	Path and action separated	A free nonempty path may evaluate to identity on the role carrier while its event record remains distinct. <i>C1, C4, C13</i> .

Claim	Disposition	Corrected scope (continued)
ET32	Cycle scope corrected	A specified directed cycle has only empty/full forward-invariant subsets. Its carrier is not its one-point orbit quotient, and that invariant topology differs from recognition topology. <i>C4, C9.</i>
ET33	Cycle scope corrected	A specified directed cycle has only empty/full forward-invariant subsets. Its carrier is not its one-point orbit quotient, and that invariant topology differs from recognition topology. <i>C4, C9.</i>
ET34	Physics claim withdrawn	Finite comparison potentials and their loop criterion are mathematical results; Maxwell, Legendre and thermodynamic identifications have not been derived. <i>C14.</i>
ET35	Path and action separated	A free nonempty path may evaluate to identity on the role carrier while its event record remains distinct. <i>C1, C4, C13.</i>
ET36	Cycle scope corrected	A specified directed cycle has only empty/full forward-invariant subsets. Its carrier is not its one-point orbit quotient, and that invariant topology differs from recognition topology. <i>C4, C9.</i>
ET37	Physics claim withdrawn	Finite comparison potentials and their loop criterion are mathematical results; Maxwell, Legendre and thermodynamic identifications have not been derived. <i>C14.</i>
ET38	Physics claim withdrawn	Finite comparison potentials and their loop criterion are mathematical results; Maxwell, Legendre and thermodynamic identifications have not been derived. <i>C14.</i>
ET39	Physics claim withdrawn	Finite comparison potentials and their loop criterion are mathematical results; Maxwell, Legendre and thermodynamic identifications have not been derived. <i>C14.</i>
ET40	Role closure corrected	Exchange/parity produce a quarter-turn on the signed role carrier. Arbitrary raw chi/kappa closure and a four-point orbit of one involution do not follow. <i>C2, C4.</i>
ET41	Seam corrected	The actual exchange fixed set is the whole signed diagonal; its projector is derived. The original zero-cost and positive-ray statements are not certified. <i>C7.</i>
ET42	Inverse scope corrected	The constructed role maps have proved inverses; arbitrary primitive acts are not declared invertible. <i>C1, C4.</i>
ET43	Role closure corrected	Exchange/parity produce a quarter-turn on the signed role carrier. Arbitrary raw chi/kappa closure and a four-point orbit of one involution do not follow. <i>C2, C4.</i>
ET44	Tower corrected	Distinct truncation maps give a strict global hierarchy; powers of an involution do not. Individual finite histories can stabilize. <i>C12.</i>
ET45	Helix claim withdrawn	Infinite records, a refinement tower and retained winding are constructed. A physical helix or torus is not forced by an involution. <i>C11, C12, C13.</i>
ET46	Mixed defect corrected	The mixed operator commutator is nonzero; every self-commutator is zero. This does not identify it with Riemannian or connection curvature. <i>C7, C14.</i>
ET47	Topology replaced	Recognition cylinders, their prefix metric and completion are constructed. They are not the upload's unsaturated image-stability topology. <i>C9, C10, C11.</i>
ET48	Universal claim rejected	The corrected constructive theory has closed written proofs. This is not a proof that the original three prose axioms force every later structure. <i>C1-C14.</i>